Brief guide to cardiac imaging for cardiology junior doctors

Viewing images and reports – Medcon and PACS

Medcon is the main viewing software for any echo or cath lab procedures. Accessible either via the desktop icon or via Hub (Home page – Clinical – Medcon MDWeb). You will need your trust login details to be activated to allow access – if delayed contact Bridget Smith (bridgetsmith@nhs.net).

PACS is the main radiology viewer for other imaging. You should have this login. If not contact the PACS office via switch board.

Transthoracic echo

There is a full clinical guideline regarding the appropriate requesting and triage of inpatient echocardiography available on Hub (link here when available). Some key points are summarised below:

1. Transthoracic echocardiography (TTE) can give valuable clinical information and is a widely requested test, but in the wrong setting, TTE may be inaccurate or delay appropriate patient care or discharge.
2. The quality of the TTE images and the report are reliant on clinical factors (e.g. body habitus, co-morbidities, setting etc) and full clinical information on a request form.
3. Inpatient TTE requests will be triaged according to national guidance on the clinical information provided by the referrer.
4. When requesting a TTE, first check one has not recently been performed and that the test needs to be completed as an inpatient.
5. Use the blue paper 'cardiology request' form and post it at cardiology reception (D template).
6. Make sure to complete the form fully with complete clinician contact details, full clinical information and the specific question to be addressed. Failure to do this will result in the request being declined.
7. Except in exceptional circumstances, all transoesophageal (TOE) requests must have already had a recent TTE performed.
8. If you need an emergency echo ask the cardiology registrar for advice or find a cardiac physiologist in D template to ask who may be free.
9. Inpatient reports will be available the same day, by paper copy and via Medcon.

Transthoracic echo with 'bubble study'

Performed to investigate intracardiac shunts (usually ASD or PFO) or occasionally AV malformation.

Dr Andrew Ludman, April 2017

Normal saline is agitated with ~1ml of air to create small bubbles. Once injected these can be visualised in the right heart and do not normally cross to the left heart unless there is a shunt.

Request must be discussed with the cardiology SpR or Consultant.

Patient will require a cannula.

Transthoracic echo with 'SonoVue contrast'

SonoVue has microbubbles which can traverse the lung capillaries and so will be seen in the left heart.

Performed to evaluate **LV function** more precisely in people with poor 'acoustic windows' or to look for **LV thrombus.**

Request must be discussed with the cardiology SpR or Consultant.

Patient will require a cannula.

Transoesophageal Echocardiogram (TOE)

Performed by passing an ultrasound probe in to the oesophagus to gain detailed imaging of the heart structures.

Particularly good for assessment of the mitral valve, possible endocarditis and to look for an intracardiac shunt.

When requesting, consider a number of factors:

- Can the patient lie flat and cooperate?
- Sedation safety – airway/jaw/spinal/respiratory problems?
- Swallowing difficulty, known oesophageal disease?
- Infection control – MRSA, HIV/Hepatitis all change the probe decontamination procedure

If uncertain, discuss with a TOE consultant.

TOE consultants: Dr M Gandhi, Dr A Taylor, Dr A Ludman, (Dr M Lovell when required).

Inpatients

Request must be discussed with the consultant of the week. Once agreed the patient is added to the inpatient whiteboard and the flow co-ordinator alerted. He/she will liaise

between the cardiac physiologist, nurse and TOE consultant to find a day/time. Patient must be consented by the ward SpR or Consultant.

Outpatients

Request must be discussed and agreed with consultant in charge of care. Once agreed a referral letter is sent to one of the TOE consultants for listing.

Procedure

Patient must be **Nil by Mouth** for at least 4 hours before procedure.

Anticoagulation continues normally. Discuss with operator if INR out of range.

Require a cannula – ideally in the right hand/arm.

Usually performed with Xylocaine local anaesthetic for the throat and conscious sedation with midazolam.

Pulse oximetry monitoring until patient fully awake.

Can offer water from 1 hour post procedure if fully awake. Food after this if no problems.

Inpatients will have a provisional result provided almost immediately that may require action by the ward team. Full report will follow on CDM or Medcon.

Outpatients will be discharged with a brief summary of the main findings prior to completion of a full report.

If received sedation then no driving for 24 hours.

Complications

Serious complications are rare. A mild sore throat is common. The most severe complication is oesophageal perforation or rupture (~1:10,000). Any significant swallowing pain or difficulty should prompt the patient to be put NBM again and discussed with the operator.

A patient information leaflet is available for additional information – if in doubt discuss with a senior colleague.

Cardiac CT

This is a CT scan which is ‘gated’ to the cardiac cycle to allow high resolution imaging of the coronary arteries. Can also be used for cardiac/aortic structure.

Performed primarily for outpatients in CT 1 in Radiology (N template) and reported by either Dr Richard Thomas, Dr Simon Harries (both Radiology) or Prof N Bellenger, Dr A Ludman (both Cardiology).

Can be requested **following a cardiology opinion** on Medway or by letter to Dr Richard Thomas (Consultant Radiologist).

Primarily used for outpatient investigation of chest pain and NICE have recently recommended that it is the first line investigation in the majority of people.

However, good images are very dependent on a number of factors:

- Being able to lie flat and hold breath for 10-15 seconds.
- A slow and regular heart rate (<70bpm at least).
- The amount of coronary calcium – the more, the worse the images.

Therefore, this test is good in stable, younger patients who can lie flat, are in sinus rhythm and have not had previous PCI procedures (we can sometimes visualise inside stents but can be tricky).

We also use it to visualise coronary artery bypass grafts or abnormal coronary anatomy.

May **sometimes be requested as an inpatient** but only after discussion with the ward cardiology consultant and then a cardiac CT consultant.

If requested as an inpatient will need a 'green' cannula insitu. If possible start a beta-blocker such as bisoprolol 2.5-5mg daily 2-3 days pre scan. If beta-blocker contraindicated then consider ivabradine 5mg twice daily.

Caution with renal dysfunction due to iodinated contrast.

If requesting as an outpatient following an admission, please make sure that a beta-blocker is provided (unless contraindicated).

Cardiac MRI

An MRI scan focussed on the heart and thoracic vasculature. Performed with ECG 'gating' and repeated breath holds and takes between 45-60mins.

Provides a large amount of information and is tailored to the clinical question – so it is very important to give as much clinical information as possible and to be clear about the question asked.

Mainly performed for outpatients in MRI 1 (N template) on a Monday/Tuesday morning (Dr A Ludman), all day Wednesday (Prof N Bellenger) and Thursday afternoon (Dr C Gibbs/Dr A Ludman). There is no planned inpatient capacity and so if needed these are fitted in as extras or cancellation slots.

The images are very dependent on:

- the patient being clinically stable and able to lie flat (in a narrow tube) for >45mins
- a controlled heart rate and rhythm (ideally sinus)
- patient being able to hold their breath repeatedly for 10-15secs

Main indications:

- Precise measurement of LV and RV volumes
- Detection of myocardial ischaemia (**Stress study**)
    - Mainly vasodilator stress with adenosine – caution in asthma
    - Can use dobutamine as well.
- Detection of myocardial infarction and whether heart muscle is alive or not (**Viability study**)
    - Used to decide about revascularisation
- Other indications:
    - Detecting LV thrombus
    - Assessment of heart valves
    - Cardiomyopathy – HCM, ARVC
    - Assessment of intracardiac shunt
    - Aortic assessment
    - Investigation of secondary causes of hypertension

Caution or Contraindicated:

- we tend to avoid giving gadolinium contrast if GFR<30ml/min unless benefit outweighs risk (small risk of nephrogenic systemic fibrosis)
- cerebral, vascular, eye or ear metallic clips or objects
- we tend to wait 6 weeks after implantation of metallic heart valves or other metallic implants unless clinically imperative. Coronary stents are fine.
- implantable loop recorders (Reveal devices) are normally fine but may cause annoying artefact sometimes and data may be erased (so make sure discussed)
- cardiac pacemakers/defibrillators are generally contraindicated unless 'conditional' devices and have been discussed with the MR consultant, pacing techs and MR radiographers

Requesting:

- Outpatient – letter to Prof N Bellenger with full clinical details and question.
- Inpatient – pink radiology request form with full details and discussed with either Prof N Bellenger or Dr A Ludman and then a copy given to Angie Pook (CMR secretary, template D)

Results:

- Inpatient scans are aimed to have a provisional report the same day and finalised within 24hours.
- Outpatient scans are generally reported the same day or within the same week.
- Available on PACS and paper report in the notes

Getting involved

During your cardiology rotation try to experience at least some Echo, cardiac CT and CMR. Ask an echocardiography technician to watch a few studies or come down to radiology to watch a CT/CMR study.

If you are interested in research or an audit of cardiac imaging then come and discuss with Dr A Ludman or Prof N Bellenger.